

22. Construct a C program to implement the Best Fit algorithm of memory management.

AIM

To write a C program to implement the **Best Fit memory allocation algorithm**, where each process is allocated to the smallest suitable memory block.

CODE

```
#include <stdio.h>

int main()
{
    int block[10], process[10];
    int nb, np;
    int i, j, best;

    printf("Enter number of blocks: ");
    scanf("%d", &nb);

    printf("Enter block sizes:\n");
    for(i = 0; i < nb; i++)
        scanf("%d", &block[i]);

    printf("Enter number of processes: ");
    scanf("%d", &np);

    printf("Enter process sizes:\n");
    for(i = 0; i < np; i++)
        scanf("%d", &process[i]);

    for(i = 0; i < np; i++)
    {
```

```

best = -1;

for(j = 0; j < nb; j++)
{
    if(block[j] >= process[i])
    {
        if(best == -1 || block[j] < block[best])
            best = j;
    }
}

if(best != -1)
{
    printf("Process %d allocated to Block %d\n",
           i + 1, best + 1);

    block[best] = block[best] - process[i];
}
else
{
    printf("Process %d cannot be allocated\n", i + 1);
}

return 0;
}

```

OUTPUT:

main.c	Run	Output
<pre>1 #include <stdio.h> 2 3 int main() 4 { 5 int block[10], process[10]; 6 int nb, np; 7 int i, j, best; 8 9 printf("Enter number of blocks: "); 10 scanf("%d", &nb); 11 12 printf("Enter block sizes:\n"); 13 for(i = 0; i < nb; i++) 14 scanf("%d", &block[i]); 15 16 printf("Enter number of processes: "); 17 scanf("%d", &np); 18 19 printf("Enter process sizes:\n"); 20 for(i = 0; i < np; i++) 21 scanf("%d", &process[i]); 22 23 for(i = 0; i < np; i++) 24 { 25 best = -1; 26 27 for(j = 0; j < nb; j++) 28 { 29 if(block[j] >= process[i]) 30 { 31 if(best == -1 block[j] < block[best]) 32 best = j; 33 } 34 } 35 36 if(best != -1)</pre>		<pre>Enter number of blocks: 2 Enter block sizes: 212 41 Enter number of processes: 5 Enter process sizes: 21 34 43 67 23 Process 1 allocated to Block 2 Process 2 allocated to Block 1 Process 3 allocated to Block 1 Process 4 allocated to Block 1 Process 5 allocated to Block 1 === CODE EXECUTION SUCCESSFUL ===</pre>

RESULT

Thus, the **Best Fit memory allocation algorithm** was successfully implemented and executed.